

SEE COMPUTER SCIENCE · CHAPTER 2

# Database Management System

Understanding how the digital world stores, organizes, and retrieves information

10 Teaching Hours · 8 Marks · SEE NEB Nepal

## 2.1 WHY THIS CHAPTER MATTERS

# Every app you use runs on this idea

- › Facebook stores **billions** of profiles
- › Your school stores marks, fees, attendance
- › Banks store every transaction you make
- › Online shopping stores products & orders

**What is the ONE system  
behind all of this?**

# Data becomes Information only when organized

### DEFINITION

**Data** — raw, unprocessed facts and figures with no meaning on their own.

**Example:** "Ram", "20", "Kathmandu"

### DEFINITION

**Information** — data that has been processed / organized to become meaningful.

**Example:** "Ram, age 20, lives in Kathmandu."

# What is a Database?

### DEFINITION

A **database** is an organized collection of related data that can be easily accessed, managed, and updated.

- › Think of it as a **digital filing cabinet**
- › Data stored in **tabular format** (rows & columns)
- › Examples: Banking systems, Airline systems, School records

# What is a DBMS?

### DEFINITION

A **Database Management System (DBMS)** is application software designed to store, manage, and organize data efficiently.



**Analogy:** If a database is a library's collection of books, the DBMS is the **librarian** — who organizes, finds, updates, and protects every book.

## 2.2 FEATURES OF DBMS

# What does a DBMS actually do?

- › **Data storage** — stores data in tables (rows & columns)
- › **Retrieval** — fetches exact data instantly using queries
- › **Update & delete** — modifies existing data safely
- › **Reduces redundancy** — avoids duplicate data entries
- › **Security** — controls who can access what

# Advantages vs Disadvantages of DBMS

### ✓ Advantages

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- › Reduces data redundancy
- › Improves data security
- › Easy & fast data retrieval
- › Multi-user access support

### ✗ Disadvantages

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- › Requires technical knowledge
- › High initial setup cost
- › Security risk if mismanaged
- › Can slow down with huge data

## 2.3 DATA TYPES

# Every column needs a Data Type

### DEFINITION

A **data type** defines what kind of value a field can store — numbers, text, dates, etc.



**Analogy:** A data type is like labelling a storage box — "Numbers only", "Text only" — so the wrong item never goes inside.



# The Data Types You Must Know

| Type         | Category  | Used for              |
|--------------|-----------|-----------------------|
| INT          | Numeric   | Whole numbers         |
| DECIMAL(p,s) | Numeric   | Exact decimal numbers |
| VARCHAR(n)   | Character | Variable-length text  |
| CHAR(n)      | Character | Fixed-length text     |
| DATE         | Date/Time | Calendar date         |
| BOOLEAN      | Logical   | True / False          |

# The Table — heart of every database

DEFINITION

A **table** stores data in rows and columns, much like a spreadsheet.

| Student_ID | Name            | Class | Date of Birth |
|------------|-----------------|-------|---------------|
| 101        | Aarosh Pokharel | 10    | 2010-03-12    |
| 102        | Subigya Nepal   | 10    | 2011-07-25    |

## 2.4 FIELD & RECORD

★★★★ High exam probability

# Field = Column | Record = Row

| Student_ID | Name            | Class | Date of Birth |
|------------|-----------------|-------|---------------|
| 101        | Aarosh Pokharel | 10    | 2010-03-12    |
| 102        | Subigya Nepal   | 10    | 2011-07-25    |

 **Field:** One column — e.g. "Name" — a single piece of information.

 **Record:** One row — all data about ONE student.

# Primary Key: The Unique ID

### DEFINITION

A **Primary Key** is a field that uniquely identifies each record. It cannot repeat and cannot be NULL.



**Analogy:** Your citizenship number — no two people share one, and everyone must have one.

# Foreign Key: The Connector Between Tables

DEFINITION

A **Foreign Key** is a field in one table that refers to the Primary Key in another table.

| BookID | BookTitle      | StudentID (FK) |
|--------|----------------|----------------|
| B_01   | Nepali History | 201            |
| B_03   | Learn Python   | 203            |

# When one field isn't enough

### DEFINITION

**Composite Key** — a primary key made by combining two or more fields.

e.g. Student\_ID + Subject → uniquely identifies one mark entry

### DEFINITION

**Candidate Key** — any field(s) that COULD be chosen as the primary key.

e.g. Roll No. and Citizenship No. are both candidates — only one becomes Primary Key

## 2.5 DATABASE RELATIONSHIPS

# Tables don't work alone — they connect

- › Relationships link tables using common fields (keys)
- › They reduce duplicate data across tables
- › Three types tested in SEE:

**One-to-One**

**One-to-Many**

**Many-to-Many**

# One-to-One

### DEFINITION

Each record in Table A relates to exactly ONE record in Table B.



**Analogy:** A person and their own citizenship number — exactly one match, always.



# One-to-Many

### DEFINITION

One record in Table A relates to MANY records in Table B.



**Analogy:** One teacher teaches many students — but each student class has one teacher.

# Many-to-Many

### DEFINITION

Many records in Table A relate to MANY records in Table B — needs a linking table.



**Analogy:** Students take many subjects, and each subject has many students.

# Meet MySQL — a Relational DBMS

- › **RDBMS** = DBMS that stores data in related tables
- › **MySQL** — free, open-source, widely used RDBMS
- › Used in banking, web apps, and mobile apps
- › Tables link using Primary & Foreign Keys

# SQL: How we "talk" to a database

### DEFINITION

A **Query** is a request sent to a database to insert, update, retrieve, or delete data.

 **Analogy:** SQL is the language you use to ASK the database questions and give it instructions.

## 2.6.2 CATEGORIES OF SQL

★★★★ High exam probability

# DDL · DML · DCL

| Category | Full Form                  | Purpose                                       |
|----------|----------------------------|-----------------------------------------------|
| DDL      | Data Definition Language   | Defines structure — CREATE, ALTER, DROP       |
| DML      | Data Manipulation Language | Manages data — INSERT, SELECT, UPDATE, DELETE |
| DCL      | Data Control Language      | Controls access & permissions                 |

# DDL: Building the Structure

| Command  | Function                               |
|----------|----------------------------------------|
| CREATE   | Creates a new table / database         |
| ALTER    | Modifies an existing table structure   |
| DROP     | Permanently deletes a table / database |
| TRUNCATE | Removes all data, keeps the structure  |

```
CREATE TABLE students (  
    ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Class INT  
);
```

# DML: Working with the Data

| Command     | Function                    |
|-------------|-----------------------------|
| INSERT INTO | Adds a new record (row)     |
| SELECT      | Retrieves data from a table |
| UPDATE      | Modifies existing records   |
| DELETE      | Removes existing records    |

```
INSERT INTO students (ID, Name, Class)
VALUES (1, 'Dikshya Pokhrel', 10);
```

# Filtering Data: WHERE and LIKE

### DEFINITION

**WHERE** — filters records that meet a condition.

```
SELECT * FROM students  
WHERE Class = 10;
```

### DEFINITION

**LIKE** — searches for a text pattern.

```
SELECT * FROM students  
WHERE Name LIKE 'A%';
```



# Reports: Presenting Data Clearly

### DEFINITION

A **Report** is an organized display of database information for analysis or printing.

- › **Summary reports** — totals & overviews
- › **Detail reports** — full record listings
- › **Grouped reports** — data organized by category

# What SEE Loves to Ask From This Chapter

- › ★★★★★ Define DBMS + one advantage 2 marks
- › ★★★★★ Define Database & DBMS + Field/Record example 4 marks
- › ★★★★★ Primary Key vs Foreign Key
- › ★★ DDL vs DML commands
- › ★★ One-to-Many relationship examples
- › ★ Composite Key / Candidate Key (MCQ trap)

# Everything in One Frame

Data → Information  
→ Database → DBMS

Table → Field (column)  
+ Record (row)

Primary Key,  
Foreign Key,  
Composite Key

1:1, 1:Many,  
Many:Many

SQL → DDL, DML, DCL

CREATE · SELECT ·  
UPDATE · DELETE

CHAPTER 2 COMPLETE

# Database Management System

Next: Chapter 3 — Multimedia